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Lubrication system and planetary gear mechanism

Abstract

A planetary gear mechanism has a helical gear. A lubrication system for the planetary gear mechanism includes a first lubricating oil supply path, a second lubricating oil supply path, and a controller. The first lubricating oil supply path is disposed on one side of the helical gear in a direction along a rotation axis of the helical gear. The first lubricating oil supply path is formed toward a tooth surface of the helical gear. The second lubricating oil supply path is disposed on an other side of the helical gear. The second lubricating oil supply path is formed toward the tooth surface of the helical gear. The controller switches discharge of lubricating oil to the tooth surface of the helical gear between the first lubricating oil supply path and the second lubricating oil supply path based on a rotation direction of the helical gear.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National stage application of International Application No. PCT/JP2023/003267, filed on Feb. 1, 2023. This U.S. National stage application claims priority under 35 U.S.C. § 119 (a) to Japanese Patent Application No. 2022-039345, filed in Japan on Mar. 14, 2022, the entire contents of which are hereby incorporated herein by reference.

BACKGROUND

Technical Field

(2) The present invention relates to a lubrication system and a planetary gear mechanism.

Background Information

(3) In recent years, high-speed rotation has been required for planetary gear mechanisms used in transmissions and the like, as the needs for electrification increase. From the perspective of reducing the noise caused by high-speed rotation and enhancing the transmission torque, helical-teeth gears are used in such a planetary gear mechanism.

(4) In the planetary gear mechanism using helical gears as shown in Japanese Patent Application Publication No. 2011-112127, the lubricating oil for lubricating the tooth surfaces flows along the rotation axis, so the lubricating oil is supplied from the upstream side in the flow direction.

SUMMARY

(5) However, in high-speed rotations, where both forward and reverse rotations are frequently used, such as in construction machinery, if lubricating oil is supplied in response to the rotation of the helical gear in one direction, there is a risk that the tooth surface is not be sufficiently lubricated when the helical gear is rotated in the opposite direction. Conversely, if lubricating oil is supplied so as to provide sufficient lubrication regardless of whether the rotor is rotated in the forward or reverse direction, the amount of lubricating oil is excessive, resulting in large agitation losses.

(6) An object of the present disclosure is to provide a lubrication system and a planetary gear mechanism that are capable of sufficiently lubricating a tooth surface and reducing agitation loss in a planetary gear mechanism using helical gears in which both forward and reverse rotations are frequently used.

(7) A lubrication system of the first aspect according to the present disclosure is a lubrication system for a planetary gear mechanism having a helical gear, and includes a first lubricating oil supply path, a second lubricating oil supply path, and a controller. The first lubricating oil supply path is disposed on one side of both sides of the helical gear in a direction along a rotation axis of the helical gear, and is formed toward the tooth surface of the helical gear. The second lubricating oil supply path is disposed on the other side of both sides of the helical gear and is formed toward the tooth surface of the helical gear. The controller switches discharge of lubricating oil to the tooth surface of the helical gear between the first lubricating oil supply path and the second lubricating oil supply path based on a rotation direction of the helical gear.

(8) A planetary gear mechanism of the second aspect according to the present disclosure includes a first lubricating oil supply path and a second lubricating oil supply path. The first lubricating oil supply path is disposed on one side of both sides of a helical gear in a direction along a rotation axis of the helical gear, and is formed toward a tooth surface of the helical gear. The second lubricating oil supply path is disposed on the other side of both sides of the helical gear and is formed toward the tooth surface of the helical gear.

(9) According to the present disclosure, it is possible to provide a lubrication system and a planetary gear mechanism that are capable of sufficiently lubricating a tooth surface and reducing agitation loss in a planetary gear mechanism using helical gears in which both forward and reverse rotations are frequently used.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram showing a drive system of a work machine according to an embodiment of the present disclosure.

(2) FIG. 2 is a block diagram showing a lubrication system according to the embodiment of the present disclosure.

(3) FIG. 3 is a cross-sectional view showing a planetary gear mechanism according to the

embodiment of the present disclosure.

(4) FIG. 4 is an enlarged diagram showing a sun gear and the vicinity thereof in FIG. 3.

(5) FIG. 5A is a diagram provided as a schematic view showing a positional relationship between an opening for discharging lubricating oil and gears when a planetary gear mechanism is viewed along a first direction.

(6) FIG. 5B is a diagram provided as a schematic view showing a positional relationship between an opening for discharging lubricating oil and gears when a planetary gear mechanism is viewed along a second direction.

(7) FIG. 6A is a diagram provided as a side schematic view showing a state in which the sun gear and a planetary gear are in mesh.

(8) FIG. 6B is a diagram provided as a schematic view of the sun gear and the planetary gear viewed along the first direction.

(9) FIG. 7 is a flow diagram showing a control operation of the lubrication system.

DETAILED DESCRIPTION OF EMBODIMENTS

(10) A planetary gear mechanism and a lubrication system having a planetary gear mechanism according to an embodiment of the present disclosure will be described with reference to the drawings. The planetary gear mechanism of this embodiment is used, for example, in the drive system of a work machine.

(11) (Structure)

(12) (Outline of Driving System 2 of Work Machine 1)

(13) FIG. 1 is a schematic diagram showing a drive system 2 of a work machine 1. The drive system 2 of the work machine 1 includes an engine 3, a torque converter 4, a transmission 5, a transfer 6, axles 7a, 7b, a pair of front tires 8, and a pair of rear tires 9. The engine 3 is, for example, a diesel engine. The driving force generated by the engine 3 is transmitted to the torque converter 4. The torque converter 4 transmits the driving force generated by engine 3 to the transmission 5.

(14) The transmission 5 decelerates the driving force transmitted from the engine 3 via torque converter 4, and transmits it to the transfer 6. The planetary gear mechanism 11 of the present embodiment, which will be described later, is used as a reducer of the transmission 5, for example.

(15) The transfer 6 distributes the driving force transmitted from the transmission 5 to the front and rear axles 7a and 7b. The pair of front tires 8 are connected to the front axle 7a. The pair of front tires 8 are rotated by the power from the engine 3 distributed to the front axle 7a. The pair of rear tires 9 are connected to the rear axle 7b. The pair of rear tires 9 are rotated by the power from the engine 3 distributed to the rear axle 7b.

(16) The work machine 1 is equipped with a lubrication system 10 for lubricating the gears of a planetary gear mechanism 11. FIG. 2 is a block diagram showing a structure of the lubrication system 10. The lubrication system 10 includes a planetary gear mechanism 11, a pump 12, a switching valve 13, a rotation sensor 14, and a controller 15.

(17) The pump 12 supplies lubricating oil to the planetary gear mechanism 11 from a tank 16 that stores lubricating oil. As will be described in detail later, the planetary gear mechanism 11 includes a plurality of supply paths for supplying lubricating oil to the gears. The switching valve 13 switches the supply of lubricating oil between the plurality of supply paths. The rotation sensor 14 detects information to determine the rotation directions of the gears. The controller 15 controls the pump 12 and the switching valve 13. The controller 15 controls the switching valve 13 based on the information detected by the rotation sensor 14.

(18) (Planetary Gear Mechanism 11)

(19) FIG. 3 is a cross-sectional diagram showing the planetary gear mechanism 11 of the present embodiment.

(20) The planetary gear mechanism 11 comprises an input shaft 21 (an example of a first shaft), a sun gear 22 (an example of a helical gear), a plurality of planetary gears 23, a pinion shaft 24, a

planetary carrier **25**, a ring gear **26** (an example of a helical gear), a fixing member **27**, an output shaft **28** (an example of a second shaft), and a housing **29**.

(21) Power is input to the input shaft **21**. In FIG. **1**, the power from the engine **3** is input via the torque converter **4**. The input shaft **21** is a cylindrical member. The input shaft **21** is positioned in the housing **29**. The input shaft **21** is rotatably supported with respect to the housing **29** by a bearing **51** that is arranged on the housing **29**. In FIG. **3**, the central axis of the input shaft **21** is shown as the line O. The input shaft **21** is arranged coaxially with the output shaft **28** which will be described later. Among the directions parallel to the central axis O, the direction extending from the input shaft **21** toward the output shaft **28** is defined as a first direction A1, and the direction extending from the output shaft **28** toward the input shaft **21** and opposite to the first direction A1 is defined as a second direction A2. The input shaft **21** includes a main body **111** and an end portion **112**. The end portion **112** is located at the end of the main body **111** in the first direction A1. FIG. **4** is an enlarged view showing the sun gear **22** and the vicinity thereof. The end portion **112** is formed to be smaller than the outer diameter of the main body **111**. The main body **111** and the end portion **112** form a step therebetween. At the step portion, the main body **111** has an end surface **111a** that is perpendicular to the central axis O.

(22) The sun gear **22** is fixed to the tip of the input shaft **21**. The sun gear **22** is arranged coaxially with the input shaft **21**. The sun gear **22** is disposed around the end portion **112**, which is disposed within the housing **29**, of the input shaft **21**. The sun gear **22** rotates about the axis O together with the input shaft **21**. The sun gear **22** is a helical gear. The sun gear **22** includes a tooth surface **22a** that has helical teeth **22b**. The helical teeth **22b** of the sun gear **22** are indicated by two-dot chain lines in FIGS. **3** and **4**. The helical teeth **22b** are arranged at an angle to the axis O.

(23) As shown in FIG. **3**, the plurality of planetary gears **23** is arranged around the sun gear **22**. In the present embodiment, three planetary gears **23** are mounted, as shown in FIG. **5A**, which will be described later. The planetary gears **23** are each a helical gear. As shown in FIG. **3**, the planetary gears **23** each include a tooth surface **23a** that has helical teeth **23b**. The helical teeth **23b** of the planetary gears **23** are indicated by two-dot chain lines in FIG. **3**. The helical teeth **23b** are arranged at an angle to the axis O. The planetary gears **23** are rotatably supported by the pinion shafts **24**. The planetary gears **23** rotate around a direction parallel to the axis O.

(24) The pinion shafts **24** are each set in the center of each planetary gear **23**. A bearing **52** is arranged around each pinion shaft **24**. The planetary gears **23** are arranged around the bearings **52**. The bearings **52** allow the planetary gears **23** to rotate against the pinion shafts **24**. The pinion shaft **24** is provided for each planetary gear **23**. In the present embodiment, three pinion shafts **24** are provided to the three planetary gears **23**, respectively.

(25) The planetary carrier **25** supports the plurality of pinion shafts **24**. The plurality of pinion shafts **24** are fixed to the planetary carrier **25**. The planetary carrier **25** includes a first carrier disk **31**, a second carrier disk **32**, and plural carrier columns **33**, a first carrier boss **34**, and a second carrier boss **35**.

(26) The first carrier disk **31** has a disk shape. First ends **241** of the plurality of pinion shafts **24** (the ends in the second direction A2) are fixed to the first carrier disk **31**. The first carrier disk **31** is disposed to the second-direction A2 side of the plurality of planetary gears **23**. The first carrier disk **31** has a through-hole therethrough along the central axis O, and the input shaft **21** is positioned in the through-hole.

(27) The second carrier disk **32** has a disk shape. Second ends **242** of the plurality of pinion shafts **24** (the ends in the first direction A1) are fixed to the second carrier disk **32**. The second carrier disk **32** is arranged to the first-direction A1 side of the plurality of planetary gears **23**. The second carrier disk **32** has a through-hole therethrough along the central axis O, and the output shaft **28** is positioned in the through-hole.

(28) The plurality of the planetary gears **23** are disposed between the first carrier disk **31** and the second carrier disk **32**. The plurality of carrier columns **33** connect between the first carrier disk **31**

and the second carrier disk **32**. The carrier columns **33** are arranged between the planetary gears **23** in a circumferential direction.

(29) The first carrier boss **34** protrudes from the first carrier disk **31** in the second direction **A2**. The input shaft **21** is positioned in the first carrier boss **34**.

(30) The second carrier boss **35** protrudes from the second carrier disk **32** in the first direction **A1**. The output shaft **28** is positioned in the second carrier boss **35**.

(31) The first carrier boss **34** is rotatably supported by the housing **29** via a bearing **53**. The second carrier boss **35** is rotatably supported by the housing **29** via a bearing **54**.

(32) The ring gear **26** is arranged around the plurality of planetary gears **23**. The ring gear **26** has an annular shape. The ring gear **26** has an inner peripheral surface where the tooth surface **26a** is formed, the tooth surface **26a** having helical teeth that mesh with the helical teeth **23b** of the planetary gears **23**. The ring gear **26** is fixed to the housing **29** via the fixing member **27**.

(33) The fixing member **27** is annular. The fixing member **27** is disposed on the outer peripheral side of the ring gear **26**. The fixing member **27** is fixed to the housing **29**. The fixing member **27** is in mesh with the teeth arranged on the outer periphery of the ring gear **26**.

(34) The output shaft **28** is arranged coaxially with the input shaft **21** (on the central axis **O**). The output shaft **28** is in mesh with the planetary carrier **25** by spline teeth. The output shaft **28** is rotatably supported with respect to the housing **29** by a plurality of bearings **55** arranged on the housing **29**. The planetary carrier **25** and the second carrier disk **32** each have a through-hole along the axis **O**. The output shaft **28** is positioned in these through-holes and fixed to the second carrier disk **32** and the second carrier boss **35**.

(35) The housing **29** accommodates the sun gear **22**, the plurality of planetary gears **23**, the pinion shafts **24**, the planetary carrier **25**, the ring gear **26**, and the fixing member **27**. The input shaft **21** and the output shaft **28** are positioned in the housing **29**.

(36) The housing **29** includes a first support portion **41**, a second support portion **42**, a third support portion **43**, and a fourth support portion **44**.

(37) The first support portion **41** is arranged to the second-direction **A2** side of the first carrier disk **31**. The first support portion **41** has a wall portion **411** and a protruding portion **412**. The wall portion **411** is arranged parallel to the first carrier disk **31**. The wall portion **411** is arranged around the first carrier boss **34**. The protruding portion **412** protrudes from the wall portion **411** in the second direction **A2**. The protruding portion **412** has a through-hole where the input shaft **21** is placed. The bearing **51** is arranged between the input shaft **21** and the inner wall of the through-hole of the protruding portion. With this configuration, the housing **29** rotatably supports the input shaft **21**. The wall portion **411** has a through-hole along the axis **O**. The first carrier boss **34** is positioned in the through-hole. The bearing **53** is arranged between the first carrier boss **34** and the inner wall of the through-hole of the wall portion **411**. With this configuration, the first support portion **41** rotatably supports the first carrier boss **34**.

(38) The second support portion **42** is arranged so as to cover the second carrier disk **32** on the first-direction **A1** side and the ring gear **26** on the radially outer side. The second support portion **42** has a wall portion **421** and an outer edge portion **422**. The wall portion **421** is arranged to the first-direction **A1** side of the second carrier disk **32**. The wall portion **421** has a through-hole along the axis **O**. The second carrier boss **35** is positioned in this through-hole. The second support portion **42** rotatably supports the second carrier boss **35**. The bearing **54** is arranged between the second carrier boss **35** and the inner wall of the through-hole of the wall portion **421**.

(39) The outer edge portion **422** extends from the outer peripheral end of the wall portion **421** in the second direction **A2**. The end of the outer edge portion **422** in the second direction **A2** is connected to the outer peripheral portion of the wall portion **411**.

(40) The first support portion **41** rotatably supports the first carrier boss **34**, and the second support portion **42** rotatably supports the second carrier boss **35**, and thereby the planetary carrier **25** is rotatably supported by the housing **29**.

(41) The third support portion **43** is disposed to the first-direction **A1** side of the second support portion **42**. The third support portion **43** is fixed to the second support portion **42**. The third support portion **43** has a through-hole along the axis **O**. The output shaft **28** is positioned in the through-hole. The plurality of bearings **55** are arranged between the inner wall of the through-hole and the output shaft **28**. With this configuration, the third support portion **43** rotatably supports the output shaft **28**.

(42) The fourth support portion **44** is disposed to the first-direction **A1** side of the third support portion **43**. The fourth support portion **44** is fixed to the third support portion **43**. The fourth support portion **44** has a through-hole along the axis **O**. The output shaft **28** is positioned in the through-hole.

(43) (Supply Paths **61-64**)

(44) Next, supply paths **61** to **64** configured to supply lubricating oil to the sun gear **22**, the planetary gears **23**, and the ring gear **26** will be described below. The planetary gear mechanism **11** further includes the supply paths **61** to **64**.

(45) The supply path **61** is arranged to the second-direction **A2** side of the sun gear **22**. The supply path **61** is configured to discharge lubricating oil toward the tooth surface **22a** of the sun gear **22**. The supply path **61** is arranged in the input shaft **21**. As shown in FIG. 4, the supply path **61** includes a first portion **611**, plural second portions **612**, and plural third portions **613**.

(46) The first portion **611** is arranged along the central axis of the main body **111** of the input shaft **21**. The plural second portions **612** extend radially outward from the end of the first portion **611** in the first direction **A1**. The third portions **613** extend in the first direction **A1** from the radially outer end of each of the second portions **612**, and have openings **61a** at the end surface **111a** of the main body **111**. The openings **61a** are arranged to face the helical teeth **22b** of the sun gear **22** in the direction along the axis **O**, as shown in FIG. 4. The openings **61a** are arranged so as to overlap the tooth surface **22a** when viewed along the first direction **A1**.

(47) FIG. 5A is a schematic diagram showing the positional relationship between the openings for discharging lubricating oil and the gears when the planetary gear mechanism **11** is viewed along the first direction **A1**. In FIG. 5A and FIG. 5B, which will be described later, the openings are shown enlarged for clarity. As shown in FIG. 5A, the plural openings **61a** of the third portions **613** are arranged along the circumferential direction around the axis **O**. The openings **61a** are arranged at equal intervals around the axis **O**. The number of openings **61a** is set to be the same as the number of the planetary gears **23**. The lubricating oil supplied from the first portion **611** in the first direction **A1** is divided into the plurality of second portions **612**, passes through the third portions **613**, and is discharged through the plural openings **61a** toward the tooth surface **22a**. The supply path **61** corresponds to an example of a first lubricating oil supply path. The opening **61a** corresponds to an example of a first discharge port.

(48) The supply path **62** is arranged to the first-direction **A1** side of the sun gear **22**. The supply path **62** is configured to discharge lubricating oil toward the tooth surface **22a** of the sun gear **22**. The supply path **62** is arranged in the output shaft **28** and the housing **29**, as shown in FIG. 3. The supply path **62** includes a first portion **621**, a second portion **622**, a third portion **623**, a fourth portion **624**, a fifth portion **625**, and a sixth portion **626**.

(49) The first portion **621** is arranged on the fourth support portion **44**. The first portion **621** extends from the outer surface of the fourth support portion **44** to the output shaft **28**. The second portion **622** is a groove formed on the outer peripheral surface of the output shaft **28** along the circumferential direction. The first portion **621** is connected to the second portion **622** which is a groove. The third portion **623** extends from the second portion **622** toward the central axis of the output shaft **28**. The fourth portion **624** extends from the end of the third portion **623** on the center side to the vicinity of the end surface **28a** of the output shaft **28** in the second direction **A2**, as shown in FIG. 4. The fifth portion **625** extends radially outward from the end of the fourth portion **624** in the second direction **A2**. The sixth portion **626** extends in the second direction **A2** from the

radially outer end of the fifth portion **625**. The sixth portion **626** has opening **62a** at the end surface **28a** of the output shaft **28** in the second direction **A2**. The opening **62a** is arranged to face the helical teeth **22b** of the sun gear **22** in the direction along the axis **O**, as shown in FIG. **4**. The opening **62a** is arranged so as to overlap the tooth surface **22a** when viewed along the second direction **A2**. FIG. **5B** is a schematic diagram showing the positional relationship between the openings for discharging lubricating oil and the gears when the planetary gear mechanism **11** is viewed along the second direction **A2**. As shown in FIG. **5B**, the opening **62a** is arranged at one location on the end surface **28a** of the output shaft **28**.

(50) The lubricating oil passes through the first portion **621**, the second portion **622**, the third portion **623**, the fourth portion **624**, the fifth portion **625**, and the sixth portion **626**, and is discharged through the opening **62a** toward the tooth surface **22a**. The supply path **62** corresponds to an example of a second lubricating oil supply path. The opening **62a** corresponds to an example of a second discharge port.

(51) The supply path **63** is arranged to the second-direction **A2** side of the ring gear **26**. The supply path **63** is configured to discharge lubricating oil toward the tooth surface **26a** of the ring gear **26**. The supply path **63** is arranged in the housing **29**. The supply path **63** is arranged on the wall portion **411** of the first support portion **41**. The supply path **63** extends from the outer surface to the inner surface of the wall portion **411**. The supply path **63** is arranged parallel to the axis **O**. The supply path **63** has an opening **63a** at the inner surface of the wall portion **411**. The opening **63a** is arranged to face the helical teeth inside the ring gear **26** in the direction along the axis **O**. The opening **63a** is arranged so as to overlap the tooth surface **26a** when viewed along the first direction **A1**. As shown in FIG. **5A**, one opening **63a** is located, for example, at the top of the tooth surface **26a** of the ring gear **26**. The supply path **63** corresponds to an example of a first lubricating oil supply path or a third lubricating oil supply path. The opening **63a** corresponds to an example of a first discharge port.

(52) The supply path **64** is arranged to the first-direction **A1** side of the ring gear **26**. The supply path **64** is configured to discharge lubricating oil toward the tooth surface **26a** of the ring gear **26**. The supply path **64** is arranged in the housing **29**. The supply path **64** is arranged on the wall portion **421** of the second support portion **42**. The supply path **64** extends from the outer surface to the inner surface of the second support portion **42**. The supply path **64** has a first portion **641** and a second portion **642**. The first portion **641** is arranged perpendicularly to the axis **O** and extends from the outer peripheral surface of the wall portion **421** toward the inside (in the direction of the axis **O**). The second portion **642** is arranged from the inner end of the first portion **641** toward the second direction **A2**. The second portion **642** has an opening **64a** at the inner peripheral surface of the wall portion **421**. The opening **64a** is arranged to face the helical teeth inside the ring gear **26** in the direction along the axis **O**. The opening **64a** faces the opening **63a** in the direction along the axis **O**. The opening **64a** is arranged so as to overlap the tooth surface **26a** when viewed along the second direction **A2**. As shown in FIG. **5B**, one opening **64a** is located, for example, at the top of the tooth surface **26a** of the ring gear **26**. The supply path **64** corresponds to an example of a second lubricating oil supply path or a forth lubricating oil supply path. The opening **64a** corresponds to an example of a second discharge port.

(53) (Switching Valve **13**)

(54) The switching valve **13** is configured to switch the supply path for supplying the lubricating oil sent from the pump **12** to the gears. The pump **12** and the tank **16** are connected by a first pipeline **71**. The switching valve **13** and the pump **12** are connected by a second pipeline **72**. A third pipeline **73** and a fourth pipeline **74** are connected to the switching valve **13**. The third pipeline **73** is connected to the supply path **61** and the supply path **63** of the planetary gear mechanism **11**. The fourth pipeline **74** is connected to the supply path **62** and the supply path **64** of the planetary gear mechanism **11**.

(55) The switching valve **13** switches the connection destination of the second pipeline **72** between

the third pipeline **73** and the fourth pipeline **74**. When the second pipeline **72** and the third pipeline **73** are connected by the switching valve **13**, lubricating oil is supplied to the supply path **61** and the supply path **63** of the planetary gear mechanism **11**. When the second pipeline **72** and the fourth pipeline **74** are connected by the switching valve **13**, lubricating oil is supplied to the supply path **62** and the supply path **64** of the planetary gear mechanism **11**.

(56) The planetary gear mechanism **11** and the tank **16** are connected by a fifth pipeline **75**. As shown in FIGS. **2** and **3**, the lubricating oil discharged from the planetary gear mechanism **11** is returned to the tank **16**.

(57) (Rotation Sensor **14**)

(58) The rotation sensor **14** detects information for determining the rotation direction of the sun gear **22**. The rotation sensor **14** transmits the detected information to the controller **15**. As the rotation sensor **14**, for example, a rotary encoder can be used.

(59) The information for determining the rotation direction of the sun gear **22** may also be the information on the rotation direction of the input shaft **21** or the output shaft **28**, or the information on the position of a forward/reverse switching lever of the work machine **1**. When the forward/reverse switching lever is in the forward position, the sun gear **22** can be determined to be rotating in a predetermined direction, and when the forward/reverse switching lever is in the reverse position, the sun gear **22** can be determined to be rotating in the opposite direction to the predetermined direction.

(60) (Controller **15**)

(61) The controller **15** includes a processor such as a CPU. The processor is configured to execute processes for controlling the pump **12** and the switching valve **13**. The controller **15** includes a storage device. The storage device includes a memory such as RAM or ROM, and an auxiliary storage device such as a hard disk drive (HDD) or a solid-state drive (SSD). The storage device stores data and programs for controlling the pump **12** and the switching valve **13**.

(62) The controller **15** controls the switching valve **13** based on the information detected by the rotation sensor **14**. Upon receiving the detected information from the rotation sensor **14**, the controller **15** determines the rotation direction of the sun gear **22**, switches the switching valve **13** according to the determined rotation direction, and causes lubricating oil to be discharged through the supply path **61** and the supply path **63**, or through the supply path **62** and supply path **64**.

(63) The rotation of helical gears and the flow of lubricating oil will be described below: FIG. **6A** is a schematic side view showing a state in which the sun gear **22** and planetary gears **23** are in mesh with each other. For easier understanding, the sizes and shapes of the planetary gears **23** are different from those in FIG. **3**. FIG. **6B** is a schematic diagram of the sun gear **22** and the planetary gears **23** when viewed along the first direction **A1**.

(64) When the sun gear **22** rotates in the left-hand rotation direction **L** shown in FIG. **6B**, the lubricating oil flows on the tooth surfaces **22a** and **23a** into the first direction **A1** shown in FIG. **6A** by the shape of the helical teeth **22b** of the sun gear **22** and the shape of the helical teeth **23b** of the planetary gears **23**. In contrast, when the sun gear **22** rotates in the right-hand rotation direction **R** shown in FIG. **6B**, the lubricating oil flows on the tooth surfaces **22a** and **23a** into the second direction **A2**.

(65) When the sun gear **22** rotates in the left-hand rotation direction **L**, the lubricating oil is supplied so as to move from the second-direction **A2** side of the sun gear **22**, which is upstream of the flowing direction, toward the first direction **A1**, which thereby enables constant supply of the lubricating oil onto the tooth surfaces **22a** and **23a**. Furthermore, when the sun gear **22** rotates in the right-hand rotation direction **R**, the lubricating oil is supplied so as to move from the first-direction **A1** side of the sun gear **22**, which is upstream of the flow direction, toward the second direction **A2**, which thereby enables constant supply of the lubricating oil onto the tooth surfaces **22a** and **23a**.

(66) Note that, between the ring gear **26** and the planetary gears **23** also, when the sun gear **22**

rotates in the left-hand rotation direction L, the lubricating oil flows in the first direction A1, and when the sun gear 22 rotates in the right-hand rotation direction R, the lubricating oil flows in the second direction A2.

(67) Therefore, when the controller 15 determines that the rotation direction of the sun gear 22 is the left-hand rotation direction L based on the information detected by the rotation sensor 14, the controller 15 drives the switching valve 13 to connect the second pipeline 72 and the third pipeline 73. As a result, the lubricating oil supplied from the tank 16 by the pump 12 is discharged toward the tooth surface 22a of the sun gear 22 through the opening 61a of the supply path 61 that is disposed to the second-direction A2 side of the sun gear 22. Further, the lubricating oil supplied from the tank 16 by the pump 12 is discharged toward the tooth surface 26a of the ring gear 26 through the opening 63a of the supply path 63 that is disposed to the second-direction A2 side of the ring gear 26.

(68) When the controller 15 determines that the rotation direction of the sun gear 22 is the right-hand rotation direction R based on the information detected by the rotation sensor 14, the controller 15 drives the switching valve 13 to connect the second pipeline 72 and the fourth pipeline 74. As a result, the lubricating oil supplied from the tank 16 by the pump 12 is discharged toward the tooth surface 22a of the sun gear 22 through the opening 62a of the supply path 62 that is disposed to the first-direction A1 side of the sun gear 22. The lubricating oil supplied from the tank 16 by the pump 12 is discharged toward the tooth surface 26a of the ring gear 26 through the opening 64a of the supply path 64 that is disposed to the first-direction A1 side of the ring gear 26.

(69) (Operation)

(70) Next, the control operation of the lubrication system 10 of the present embodiment will be described. FIG. 7 is a flow diagram showing the control operation of the lubrication system 10 of the present embodiment.

(71) First, in Step S101, the controller 15 drives the pump 12. Next, in Step S102, the controller 15 receives information detected by the rotation sensor 14.

(72) In Step S103, then the controller 15 determines the rotation direction of the sun gear 22 based on the received detection information. For example, the controller 15 determines whether the rotation direction of the sun gear 22 is the left-hand rotation direction L, based on the detection information.

(73) In Step S103, when it is determined that the rotation direction is the left-hand rotation direction L, the control proceeds to Step S104.

(74) In Step S104, the controller 15 drives the switching valve 13 to connect the second pipeline 72 and the third pipeline 73, and the control ends. As a result, the lubricating oil supplied from the tank 16 by driving the pump 12 is discharged toward the tooth surface 22a of the sun gear 22 through the opening 61a of the supply path 61 that is located to the second-direction A2 side of the sun gear 22. Further, the lubricating oil supplied from the tank 16 by driving the pump 12 is discharged toward the tooth surface 26a of the ring gear 26 through the opening 63a of the supply path 63 that is disposed to the second-direction A2 side of the ring gear 26.

(75) In contrast, in Step S103, when it is determined that the rotation direction is not the left-hand rotation direction L, the rotation direction can be determined to be the right-hand rotation direction R, and the control proceeds to Step S105.

(76) In Step S105, the controller 15 drives the switching valve 13 to connect the second pipeline 72 and the fourth pipeline 74, and the control ends. As a result, the lubricating oil supplied from the tank 16 by driving the pump 12 is discharged toward the tooth surface 22a of the sun gear 22 through the opening 62a of the supply path 62 that is disposed to the first-direction A1 side of the sun gear 22. The lubricating oil supplied from the tank 16 by driving the pump 12 is discharged toward the tooth surface 26a of the ring gear 26 through the opening 64a of the supply path 64 that is disposed to the first-direction A1 side of the ring gear 26.

(77) Note that the rotation sensor 14 constantly detects information regarding the rotation direction

of the sun gear **22** and sends it to the controller **15**. Therefore, the rotation direction is determined every time detection information is received. For example, when the rotation direction of the sun gear **22** determined based on the previous detection information matches the rotation direction of the sun gear **22** determined based on the present detection information, the controller **15** is able to maintain the current state, without driving the switching valve **13**.

(78) (Features and Others)

(79) (1)

(80) The lubrication system **10** of the present embodiment is the lubrication system **10** for the planetary gear mechanism **11** including a sun gear **22** (an example of a helical gear) having helical teeth **22b**, and includes the supply path **61** (an example of a first lubricating oil supply path), a supply path **62** (an example of a second lubricating oil supply path), and the controller **15**. The supply path **61** is disposed on one side of both sides of the sun gear **22** in the direction along the rotation axis O of the sun gear **22** and is formed toward the tooth surface **22a** of the sun gear **22**. The supply path **62** is disposed on the other side of the both sides of the sun gear **22** and is formed toward the tooth surface **22a** of the sun gear **22**. The controller **15** switches the discharge of lubricating oil to the tooth surface **22a** of the sun gear **22** between the supply path **61** and the supply path **62** based on the rotation direction of the sun gear **22**.

(81) By switching between the supply path **61** and the supply path **62** in accordance with the rotation of the sun gear **22**, an appropriate amount of lubricating oil can be supplied from upstream of the direction in which the lubricating oil flows by the sun gear **22**, regardless of the direction in which sun gear **22** rotates. This allows the tooth surface **22a** of the sun gear **22** to be sufficiently lubricated, thereby reducing stirring loss.

(82) (2)

(83) In the lubrication system **10** of the present embodiment, the supply path **61** has the opening **61a** (an example of a first discharge port). The opening **61a** is arranged so as to overlap with the tooth surface **22a** of the sun gear **22** when viewed from the direction along the rotation axis O, and the lubricating oil is discharged through the opening **61a**. The supply path **62** has the opening **62a** (an example of a second discharge port). The opening **62a** is arranged so as to overlap with the tooth surface **22a** of the sun gear **22** when viewed from the direction along the rotation axis O, and the lubricating oil is discharged through the opening **62a**.

(84) As a result, it is possible to lubricate the tooth surface **22a** of the sun gear **22** by the lubricating oil discharged from the opening **61a** or the opening **62a**.

(85) (3)

(86) The lubrication system **10** of the present embodiment further includes the input shaft **21** (an example of a first shaft) and a plurality of planetary gears **23**. The sun gear **22** is fixed to the input shaft **21**. The plurality of planetary gears **23** are arranged around the sun gear **22** and mesh with the sun gear **22**. The supply path **61** has a plurality of openings **61a** through which the lubricating oil is discharged. The plurality of openings **61a** are arranged in the input shaft **21**.

(87) As a result, when the sun gear **22** rotates in the left-hand rotation direction L, it is possible to lubricate the space between the sun gear **22** and the planetary gears **23** by the lubricating oil.

(88) (4)

(89) In the lubrication system **10** of the present embodiment, the plurality of openings **61a** are arranged at equal intervals in the circumferential direction. The number of the openings **61a** is equal to or greater than the number of the planetary gears **23**. As a result, it is possible to supply a sufficient amount of lubricating oil between the sun gear **22** and the plurality of planetary gears **23**.

(90) (5)

(91) The lubrication system **10** of the present embodiment further includes the planetary carrier **25**, the housing **29**, and the output shaft **28** (an example of a second shaft). The planetary carrier **25** rotatably supports the plurality of planetary gears **23**. The housing **29** rotatably supports the planetary carrier **25**. The output shaft **28** is fixed to the planetary carrier **25** and is arranged

coaxially with the input shaft **21**. The supply path **62** has the opening **62a** through which the lubricating oil is discharged. The opening **62a** is arranged in the output shaft **28**.

(92) As a result, when the sun gear **22** rotates in the right-hand rotation direction R, it is possible to supply lubricating oil between the sun gear **22** and the planetary gears **23**.

(93) (6)

(94) The lubrication system **10** of the present embodiment further includes the plurality of planetary gears **23**, the ring gear **26**, the supply path **63** (an example of a third lubricating oil supply path), and the supply path **64** (an example of a fourth lubricating oil supply path). The plurality of planetary gears **23** are arranged around the sun gear **22** and mesh with the tooth surface **22a** of the sun gear **22**. The ring gear **26** is arranged around the plurality of planetary gears **23** and meshes with the plurality of planetary gears **23**. The supply path **63** is disposed on one side of both sides of the ring gear **26** in the direction along the rotation axis O, and is formed toward the tooth surface **26a** of the ring gear **26**. The supply path **64** is disposed on the other side of the both sides of the ring gear **26** and is formed toward the tooth surface **26a** of the ring gear **26**.

(95) As a result, it is possible to supply an appropriate amount of lubricating oil between the sun gear **22** and the plurality of planetary gears **23**, and between the plurality of planetary gears **23** and the ring gear **26**, regardless of whether the sun gear **22** rotates in the left-hand rotation direction L or in the right-hand rotation direction R.

(96) (7)

(97) The lubrication system **10** of the present embodiment is the lubrication system **10** for the planetary gear mechanism **11** including a ring gear **26** (an example of a helical gear) having helical teeth, and includes the supply path **63** (an example of a first lubricating oil supply path), the supply path **64** (an example of a second lubricating oil supply path), and the controller **15**. The supply path **63** is disposed on one side of both sides of the ring gear **26** in the direction along the rotation axis O of the ring gear **26**, and is formed toward the tooth surface **26a** of the ring gear **26**. The supply path **64** is disposed on the other side of the two sides of the ring gear **26** and is formed toward the tooth surface **26a** of the ring gear **26**. The controller **15** switches the discharge of lubricating oil to the tooth surface **26a** of the ring gear **26** between the supply path **63** and the supply path **64** based on the rotation direction of the ring gear **26**.

(98) By switching between the supply path **63** and the supply path **64** in accordance with the rotation of ring gear **26**, lubricating oil can be supplied from the upstream side of the direction in which the lubricating oil flows by the ring gear **26**, regardless of the direction in which the ring gear **26** rotates. This allows the tooth surface **26a** of the ring gear **26** to be sufficiently lubricated, thereby reducing stirring loss.

(99) (8)

(100) In the lubrication system **10** of the present embodiment, the supply path **63** has the opening **63a** (an example of a first discharge port) arranged so as to overlap with the tooth surface **26a** of the ring gear **26** when viewed from a direction along the rotation axis O, and the lubricating oil is discharged through the opening **63a**. The supply path **64** has the opening **64a** (an example of a second discharge port) arranged so as to overlap with the tooth surface **26a** of the ring gear **26** when viewed from the direction along the rotation axis O, and the lubricating oil is discharged through the opening **64a**.

(101) As a result, it is possible to lubricate the tooth surface **26a** of the ring gear **26** by the lubricating oil discharged from the opening **63a** or the opening **64a**.

(102) (9)

(103) The lubrication system **10** of the present embodiment further includes the sun gear **22**, the plurality of planetary gears **23**, the planetary carrier **25**, and the housing **29**. The plurality of planetary gears **23** are arranged around the sun gear **22** and mesh with the sun gear **22**. The planetary carrier **25** rotatably supports the plurality of planetary gears **23**. The ring gear **26** (an example of a helical gear) is arranged around the plurality of planetary gears **23** and meshes with

the plurality of planetary gears **23**. Each of the supply path **63** and the supply path **64** is arranged in the housing **29**.

(104) As a result, it is possible to lubricate sufficiently between the ring gear **26** and the planetary gear **23**.

(105) (10)

(106) The planetary gear mechanism **11** of the present embodiment includes the supply path **61** (an example of a first lubricating oil supply path) and the supply path **62** (an example of a second lubricating oil supply path). The supply path **61** is disposed on one side of both sides of the sun gear **22** (an example of a helical gear) in a direction along the rotation axis O of the sun gear **22**, and is formed toward the tooth surface **22a** of the sun gear **22**. The supply path **62** is disposed on the other side of the both sides of the sun gear **22** and is formed toward the tooth surface **22a** of the sun gear **22**.

(107) By discharging lubricating oil from either the supply path **61** or the supply path **62** in accordance with the rotation of the sun gear **22**, lubricating oil can be supplied from the upstream side of the direction in which the lubricating oil flows by the sun gear **22**, regardless of the direction in which sun gear **22** rotates. This allows the tooth surface **22a** of the sun gear **22** to be sufficiently lubricated, thereby reducing stirring loss.

(108) (11)

(109) The planetary gear mechanism **11** of the present embodiment includes the supply path **63** (an example of a first lubricating oil supply path) and the supply path **64** (an example of a second lubricating oil supply path). The supply path **63** is disposed on one side of both sides of the ring gear **26** (an example of a helical gear) in the direction along the rotation axis O of the ring gear **26** and is formed toward the tooth surface **26a** of the ring gear **26**. The supply path **64** is disposed on the other side of the both sides of the ring gear **26** and is formed toward the tooth surface **26a** of the ring gear **26**.

(110) By discharging lubricating oil from either the supply path **63** or the supply path **64** in accordance with the rotation of the ring gear **26**, lubricating oil can be supplied from the upstream side of the direction in which the lubricating oil flows by the ring gear **26**, regardless of the direction in which ring gear **26** rotates. This allows the tooth surface **26a** of the ring gear **26** to be sufficiently lubricated, thereby reducing stirring loss.

OTHER EMBODIMENTS

(111) One embodiment of the claimed invention has been described above. However, the claimed invention is not limited to the above embodiment, and various changes can be made without departing from the scope of the claimed invention. In particular, the multiple embodiments and modifications described in this specification can be arbitrarily combined as necessary.

(112) (A)

(113) In the above embodiment, the supply path **61** has three openings **61a**, but the claimed invention is not limited thereto, and the number may be four or more, or two or less.

(114) Although the supply path **62** has only one opening **62a**, the claimed invention is not limited thereto and may have a plurality of openings **62a**.

(115) (B)

(116) In the above embodiment, the lubricating oil is discharged to the tooth surface **26a** of the ring gear **26** from one opening **63a** on the first direction A1 side, but the lubricating oil may be discharged from a plurality of openings. Furthermore, the lubricating oil is discharged to the tooth surface **26a** of the ring gear **26** from one opening **64a** on the second direction A2 side, but the lubricating oil may be discharged from a plurality of openings.

(117) (C)

(118) In the above embodiment, the opening **61a** and the opening **62a** are disposed to overlap with the tooth surface **22a** of the sun gear **22** when viewed along the axis O, but the claimed invention is not limited to this. It is only necessary that the lubricating oil discharged from the opening **61a** and

the opening **62a** is supplied to the tooth surface **22a**.

(119) Furthermore, the opening **63a** and the opening **64a** are arranged so as to overlap with the tooth surface **26a** of the ring gear **26** when viewed along the axis O, but this is not limited to this. It is only necessary that the lubricating oil discharged from the openings **63a** and **64a** is supplied to the tooth surface **26a**.

(120) (D)

(121) In the above embodiment, the ring gear **26** is fixed to the housing **29**, the sun gear **22** and the planetary gears **23** rotate relative to the housing **29**, and power is input to the sun gear **22** and output from the planetary carrier **25**. However, the claimed invention is not limited to this configuration. For example, a configuration is possible in which the planetary carrier **25** is fixed to the housing **29**, the sun gear **22** and the ring gear **26** rotate, and power is output from the ring gear.

(122) (E)

(123) In the planetary gear mechanism **11** of the above embodiment, three planetary gears **23** are arranged, but the number is not limited to three, and the number may be four or more.

(124) (F)

(125) The work machine **1** of the above embodiment may be a wheel loader and a forklift equipped with front and rear tires, for example, but are not limited thereto, and may be an excavator and a bulldozer, and the like that are equipped with crawler belts.

(126) The lubrication system and the planetary gear mechanism of the present disclosure are effective in lubricating the tooth surfaces sufficiently and reducing stirring loss.

Claims

1. A lubrication system for a planetary gear mechanism having a helical gear, the lubrication system comprising: a first lubricating oil supply path disposed on one side of the helical gear in a direction along a rotation axis of the helical gear, the first lubricating oil supply path being formed toward a tooth surface of the helical gear; a second lubricating oil supply path disposed on an other side of the helical gear, the second lubricating oil supply path being formed toward the tooth surface of the helical gear; and a controller configured to switch discharge of lubricating oil to the tooth surface of the helical gear between the first lubricating oil supply path and the second lubricating oil supply path based on a rotation direction of the helical gear.

2. The lubrication system according to claim 1, wherein the first lubricating oil supply path includes a first discharge port from which lubricating oil is discharged, the first discharge port is arranged so as to overlap with the tooth surface of the helical gear when viewed from the direction along the rotation axis, the second lubricating oil supply path includes a second discharge port from which lubricating oil is discharged, and the second discharge port is arranged so as to overlap with the tooth surface of the helical gear when viewed from the direction along the rotation axis.

3. The lubrication system according to claim 1, further comprising: a first shaft to which the helical gear is fixed as a sun gear; and a plurality of planetary gears arranged around the helical gear, the plurality of planetary gears meshing with the tooth surface of the helical gear, the first lubricating oil supply path including a plurality of first discharge ports from which lubricating oil is discharged, and the plurality of first discharge ports being arranged in the first shaft.

4. The lubrication system according to claim 3, wherein the first discharge ports are arranged at equal intervals in a circumferential direction, and a number of the first discharge ports is equal to or greater than a number of the planetary gears.

5. The lubrication system according to claim 3, further comprising: a planetary carrier rotatably supporting the plurality of planetary gears; a housing rotatably supporting the planetary carrier; and a second shaft fixed to the planetary carrier, the second shaft being arranged coaxially with the first shaft, the second lubricating oil supply path including a second discharge port from which lubricating oil is discharged, and the second discharge port being arranged in the second shaft.

6. The lubrication system according to claim 1, further comprising: a plurality of planetary gears arranged around the helical gear serving as a sun gear, the plurality of planetary gears meshing with the tooth surface of the helical gear; a ring gear arranged around the plurality of planetary gears, the ring gear meshing with the plurality of planetary gears; a third lubricating oil supply path disposed on the one side of the ring gear in the direction along the rotation axis, the third lubricating oil supply path being formed toward a tooth surface of the ring gear; and a fourth lubricating oil supply path disposed on the other side of the ring gear, the fourth lubricating oil supply path being formed toward the tooth surface of the ring gear, the controller being configured to switch discharge of lubricating oil to the tooth surface of the helical gear between the third lubricating oil supply path and the fourth lubricating oil supply path based on the rotation direction of the helical gear.

7. The lubrication system according to claim 1, further comprising: a sun gear; a plurality of planetary gears arranged around the sun gear, the plurality of planetary gears meshing with the sun gear; a planetary carrier rotatably supporting the plurality of planetary gears; and a housing rotatably supporting the planetary carrier, the helical gear being arranged around the plurality of planetary gears and meshing with the plurality of planetary gears, and each of the first lubricating oil supply path and the second lubricating oil supply path being arranged in the housing.

8. A planetary gear mechanism having a helical gear, the planetary gear mechanism comprising: a first lubricating oil supply path disposed on one side of the helical gear in a direction along a rotation axis of the helical gear, the first lubricating oil supply path being formed toward a tooth surface of the helical gear; and a second lubricating oil supply path disposed on an other side of the helical gear, the second lubricating oil supply path being formed toward the tooth surface of the helical gear, the first lubricating oil supply path including a first discharge port from which lubricating oil is discharged, the first discharge port being arranged so as to overlap with the tooth surface of the helical gear when viewed from the direction along the rotation axis, the second lubricating oil supply path including a second discharge port from which lubricating oil is discharged, and the second discharge port being arranged so as to overlap with the tooth surface of the helical gear when viewed from the direction along the rotation axis, the first discharge port facing toward the tooth surface of the helical gear along the rotation axis, and the second discharge port facing toward the tooth surface of the helical gear along the rotation axis.
